TD 1

Exercise 1

A company wants to develop an online shopping platform that must support 50,000 simultaneous users. The platform includes a web interface, order processing, payment services, and a database.

1. Propose a suitable cloud architecture by identifying the Front End, Back End, Business Logic, and Data Layer.
2. Which cloud services would you use to store product images and customer information? Justify your choices.
3. Explain how the architecture can remain available if one server fails.
4. Should the company use a monolithic or microservices architecture? Justify your answer.

Solution Exercise 1

1. Suitable cloud architecture is **N-Tiers** :
   - **Front End** : web interface
   - **Business Logic** :order processing and payment services
   - **Data Layer** : database
2. Cloud services :
   - **Customer Data** : RDS because structured
   - **Images** : S3 because not structured
3. We can use replication servers and backups.
4. the company should use microservices architecture for scalability, resilience, better maintenance & loose coupling because we have multiple services.

Exercise 2

A university owns one powerful physical server but needs to provide separate environments for three departments: Computer Science, Mathematics, and Electronics.

1. Explain how virtualization can solve this problem.
2. How many virtual machines would you propose, and why?
3. What is the role of the hypervisor in this architecture?
4. If one VM consumes too much CPU, how could this affect the other VMs?
5. Explain the difference between VM migration and VM replication.

Solution Exercise 2

1. Virtualization offers isolation between **VMs** and is cost-effective we can have multiple **VMs** in one host server(one physical machine)
2. We need 3 **VMs** one for each department (isolated from each other)
3. To create and manage the **VMs**
4. Reduced performance of the other **VMs** because it is consuming most of the availble host resources (resource contention)
5. The difference :
   - **VM migration :** moving a **VM** from one host machine to another one
   - **VM replication :** copying a **VM** from to another host machine while keeping the copy synchronized

Exercise 3

A hospital uses a cloud platform to store medical documents, patient images, and administrative files. Different computers running Windows and Linux must access these files.

1. Which storage solution would you recommend for sharing files between different operating systems? Justify your answer.
2. Explain the roles of **SMB** and **NFS** in this architecture.
3. What is the role of a Virtual File System (**VFS**)?
4. Should the hospital use object storage or block storage for medical images? Justify your answer.
5. Propose two mechanisms to protect stored data against unauthorized access and data loss.

Solution Exercise 3

1. File virtualization because it abstracts the file storage from the physical devices & location (bridge between cloud interface and physical file)
2. The roles :
   - **SMB** : file sharing protocole used in windows
   - **NFS** : file sharing protocole used in linux
3. The role of the **VFS** is to provide a unified access without changing the access path
4. the hospital should use object storage for medical images because they are not structured
5. We can use firewalls , Authentication and encryption to protect stored data against unauthorized access and data loss.

Exercise 4

A photo-sharing application allows users to upload thousands of images every hour.

1. Explain how **AWS** Lambda can automatically process uploaded images.
2. Identify the event that triggers the function.
3. Explain why **FaaS** is suitable for this application.
4. Compare the Pay-as-you-go model of serverless computing with traditional server hosting.
5. Identify one limitation of serverless computing and propose a solution.

Solution Exercise 4

1. **AWS** Lambda can automatically process uploaded images when the event is triggered.
2. The event that triggers the function is user uploading images e.g : clicking on upload button.
3. **FaaS** is suitable for this application we only need to process when the event is triggered and we don't have to manage the cloud architecture
4. Comparison :
   - **Pay-as-you-go :** we only pay for the code execution time
   - **Traditional Hosting :** we pay for the up time of the server even when unused idle
5. One limitation of serverless computing is **cold start latency** having to load the execution environment after being idle to answer the request to fix this we can use **provisioned concurrency allocation** to pre-initialize the execution environment.

Exercise 5

A bank develops an application using three independent services: customer management (Java), payment processing (Python), and notification services (Node.js).

1. Explain why this architecture is considered Service-Oriented Architecture (**SOA**).
2. Why are these services described as loosely coupled?
3. Explain how interoperability allows these services to communicate despite using different programming languages.
4. Compare **SOAP** and **REST** for communication between these services.
5. What happens if the notification service becomes unavailable? Does it necessarily stop the payment service? Justify your answer.

Solution Exercise 5

1. Because we have multiple, independent and interoperable services
2. These services are described as loosely coupled because they are independent from each other
3. Interoperability allows these services to communicate despite using different programming languages by using a unified interface for communication **API**
4. Comparison :
   - **SOAP** : rigid , uses **XML** , used in enterprise application
   - **REST** : flexible, uses **JSON**, used in web application
5. No it doesn't affect other services because they are independent from each other